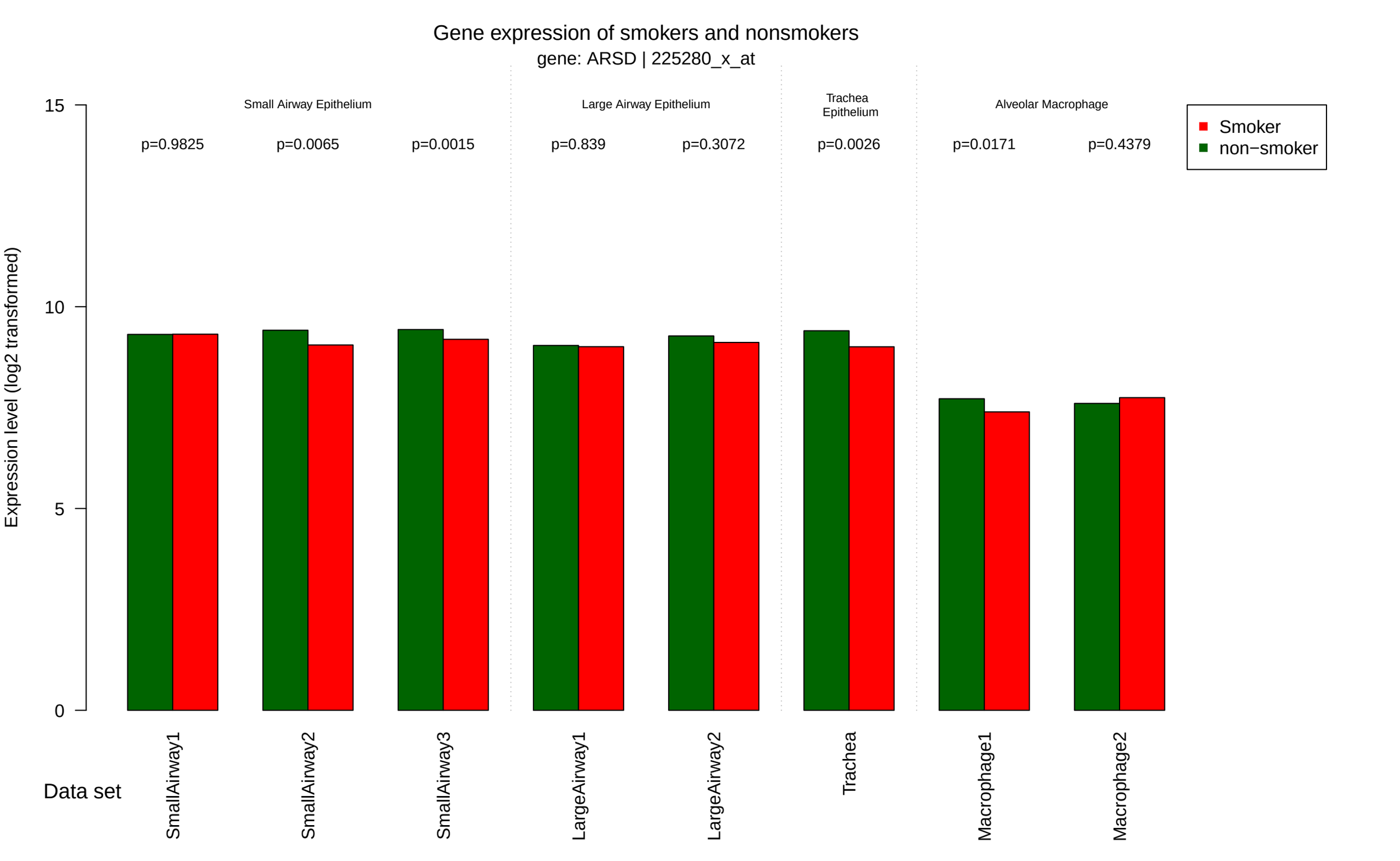
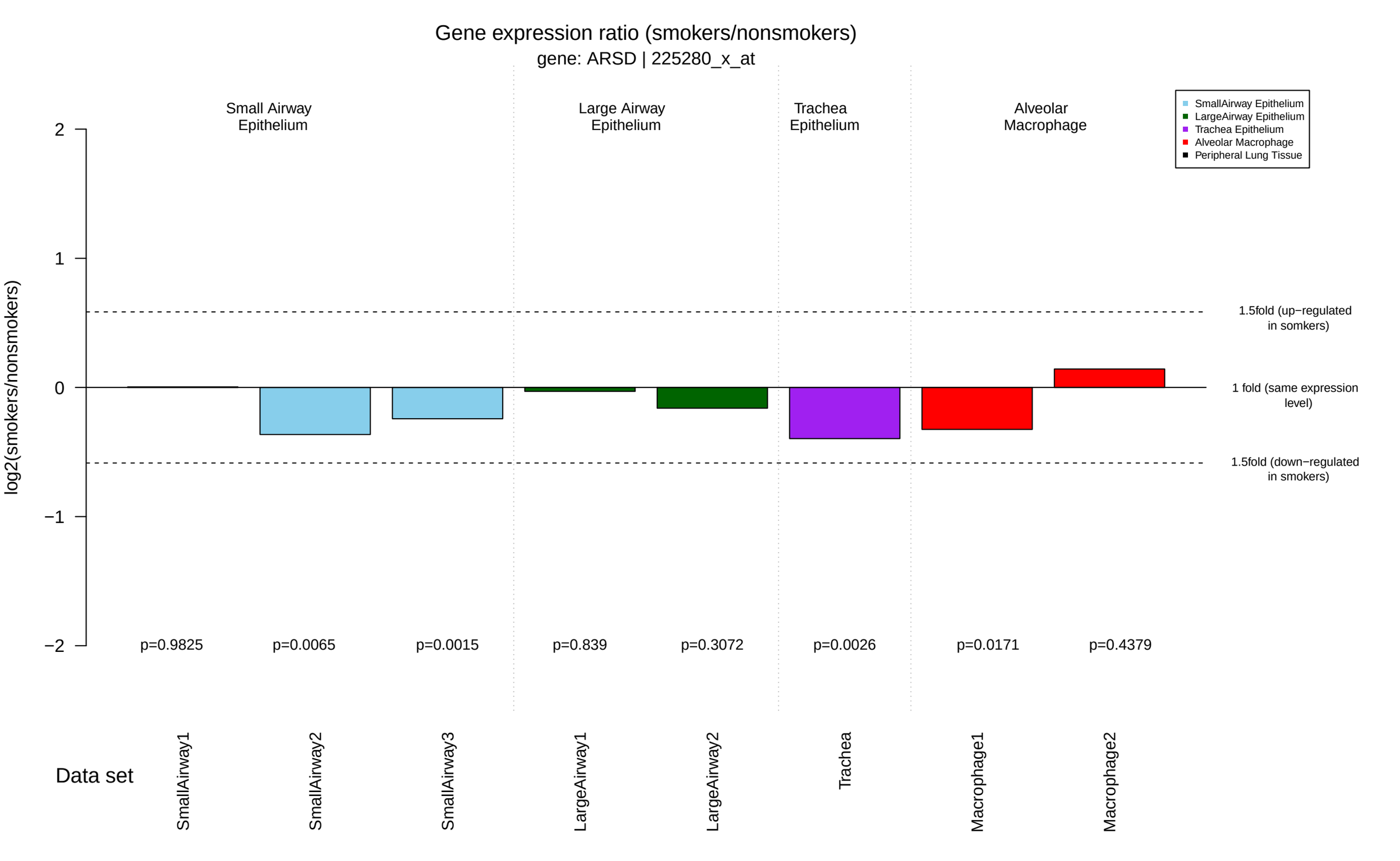
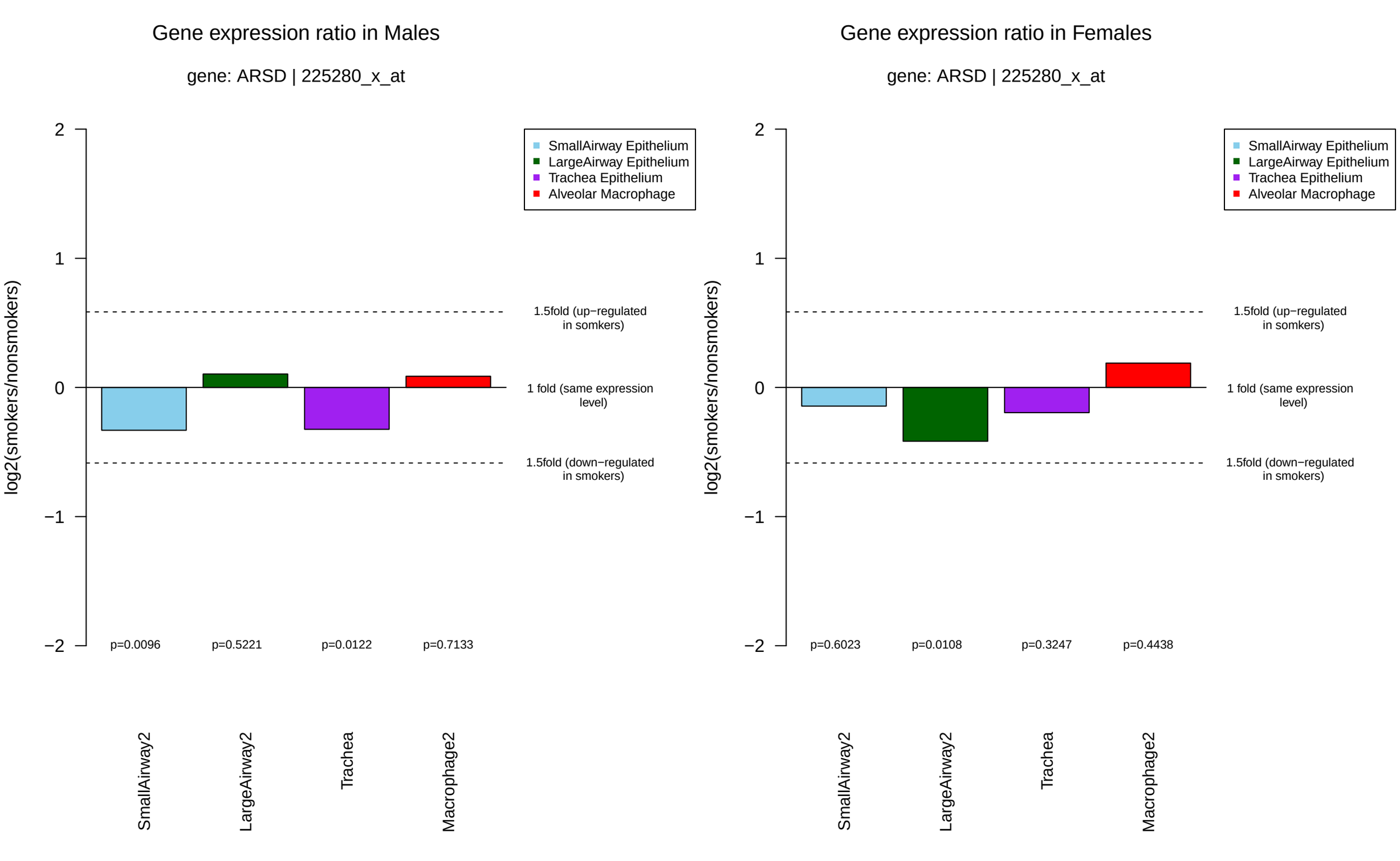
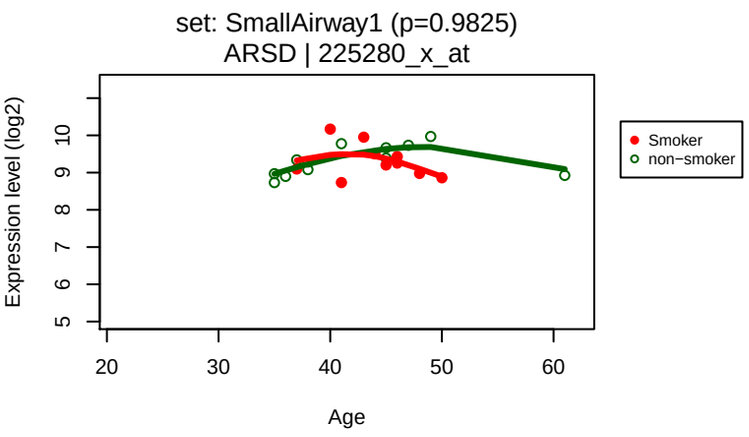
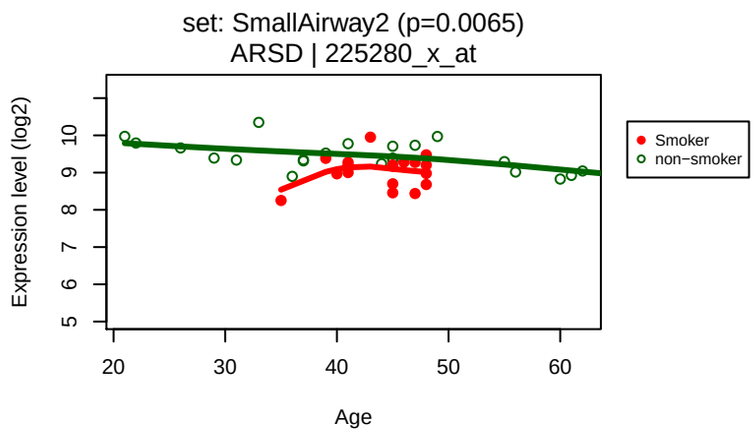
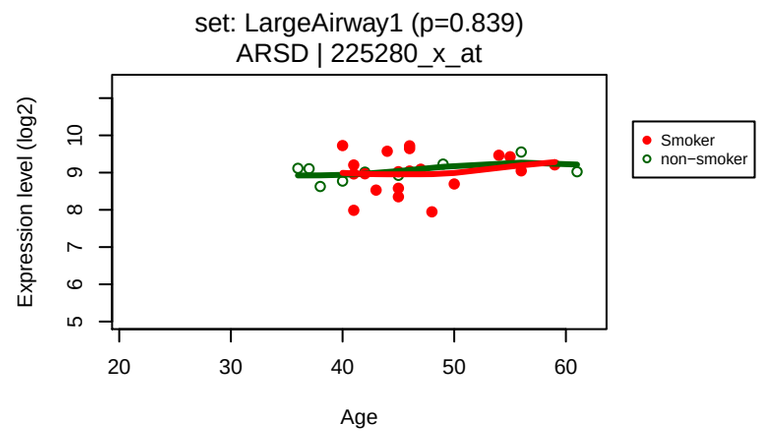

Gene expression of smokers and nonsmokers
gene: ARSD | 225280_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9825
p=0.0065
p=0.0015
p=0.839
p=0.3072
p=0.0026
p=0.0171
p=0.4379
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

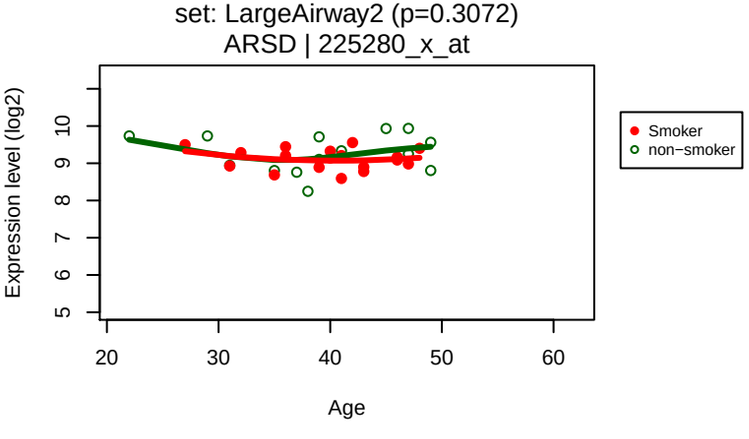

Gene expression ratio (smokers/nonsmokers)
gene: ARSD | 225280_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9825
p=0.0065
p=0.0015
p=0.839
p=0.3072
p=0.0026
p=0.0171
p=0.4379
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

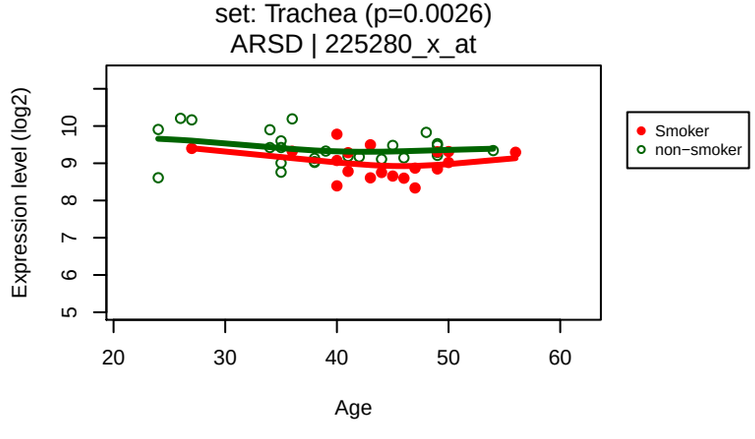

Gene expression ratio in Males
gene: ARSD | 225280_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0096
p=0.5221
p=0.0122
p=0.7133
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ARSD | 225280_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6023
p=0.0108
p=0.3247
p=0.4438
SmallAirway2
LargeAirway2
Trachea
Macrophage2

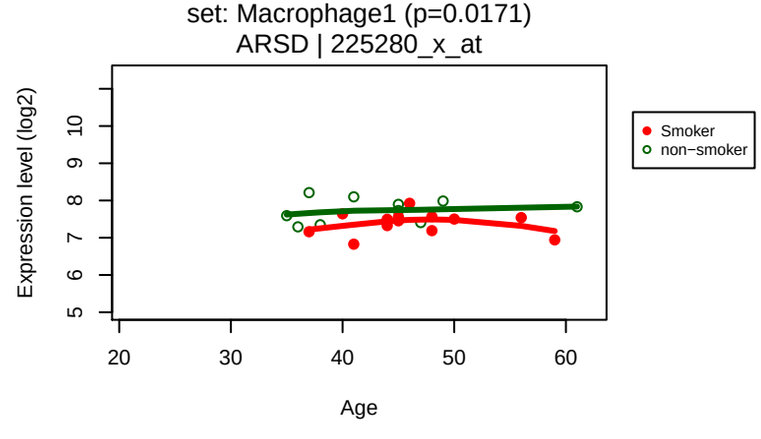

set: SmallAirway1 (p=0.9825)
ARSD | 225280_x_at
Expression level (log2)
Age
Smoker
non−smoker

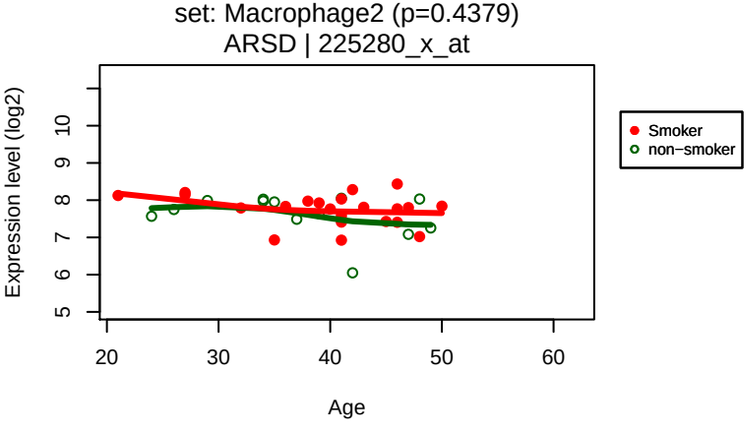

set: SmallAirway2 (p=0.0065)
ARSD | 225280_x_at
Expression level (log2)
Age
Smoker
non−smoker

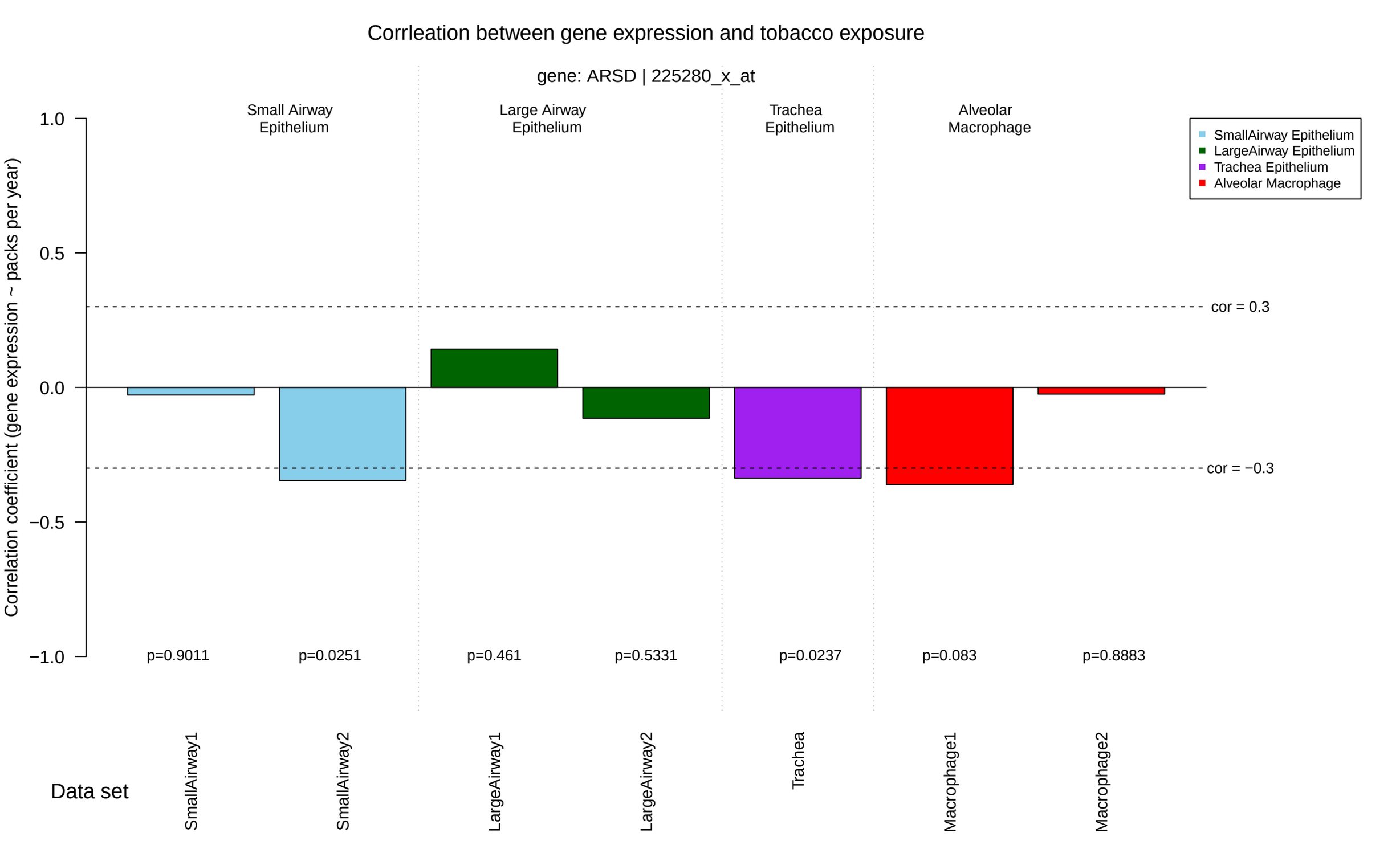

set: LargeAirway1 (p=0.839)
ARSD | 225280_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3072)
ARSD | 225280_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0026)
ARSD | 225280_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0171)
ARSD | 225280_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4379)
ARSD | 225280_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ARSD | 225280_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9011
p=0.0251
p=0.461
p=0.5331
p=0.0237
p=0.083
p=0.8883
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2